Some 'tree' facts

In the last column, we discussed the use of trees. I will now list some of their properties that you can employ while designing the strategy:

- There will only be one node that connects two nodes in a tree
- If a tree has N nodes, there will be N-1 edges
- For any binary tree with N internal nodes, there are N+1 external nodes
- The height of a given full binary tree with N internal nodes is about log N / log 2

Some evolutionary concepts: In a nutshell

Evolutionary algorithms themselves form another major branch. We will confine ourselves to some basic ideas, problem definitions and generalisations (definitions).

General single-objective optimisation problem: This is defined as minimising (or maximising) *f (x) subject to gi (x) ≤ 0, i = {1, . . . , m}, and hj (x) = 0, j = {1, . . . , p} x ∈ Ω. A solution minimises (or maximises) the scalar f (x) where x is a n-dimensional decision variable vector x = (x1 , . . . , xn) from some universe Ω.*

Single-objective global minimum optimisation: *Given a function f : Ω ⊆ Rn → R, Ω = ø, for x ∈ Ω the value f * f (x*) > -∞ is called a global minimum if and only* if

$$\forall x \in \Omega : f(x^*) \leq f(x)$$

x is by definition the global minimum solution, f is the objective function, and the set Ω is the feasible region of x.*

Useful facts:

- The purpose of finding the global minimum solution(s) is called the global optimisation problem for a single-objective problem.
- Evolutionary multi-objective optimisation (EMO) refers to the use of evolutionary algorithms of any sort (like genetic algorithms, evolution strategies, evolutionary programming or genetic programming) to solve multi-objective optimisation problems.
- Other meta-heuristics that are being used to solve multi-objective optimisation problems include particle swarm optimisation, artificial immune systems and cultural algorithms.
- Differential evolution, ant colony, tabu search, scatter search, and memetic algorithms are other key ideas in the realm.

Key ideas:

- You must see that non-dominated points are preserved in objective space, and the associated solution points in the decision space.
- The design should be such that it should continue to allow algorithmic progress towards the Pareto Front in the objective function space.
- Maintain the diversity of points on Pareto/phenotype front (space) or of Pareto optimal solutions on decision/ genotype space.
- Provide the decision maker (DM) sufficient but limited number of Pareto points for the selection (which results in decision variable values).

Please let me know if you wish to discuss these ideas more in depth.

```
 }
}

Shape {
 appearance Appearance {
  material Material {
   emissiveColor 0 1 0
   transparency 0
  }
 }
 geometry IndexedLineSet {
  coord Coordinate {
   point [
    0 707.106781186547 707.106781186548,
    353.553390593274 707.106781186547 612.372435695795,
    612.372435695795 707.106781186547 353.553390593274,
    707.106781186548 707.106781186547 4.32963728535968e-14,
    612.372435695795 707.106781186547 -353.553390593274,
    353.553390593274 707.106781186547 -612.372435695795,
    8.65927457071935e-14 707.106781186547 -707.106781186548,
    -353.553390593274 707.106781186547 -612.372435695795,
    -612.372435695794 707.106781186547 -353.553390593274,
    -707.106781186548 707.106781186547 -1.2988911856079e-13,
    -612.372435695795 707.106781186547 353.553390593274,
    -353.553390593274 707.106781186547 612.372435695794,
    -1.73185491414387e-13 707.106781186547 707.106781186548
   ]
  }
  coordIndex [ 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12 ]
 }
}
.........
..................
```

This way the code goes on. (The complete code of flatgid.wrl is available at *aasisvinayak.com/new_zone/forum.php?do=viewtopic&cat=2&topic=1*)

We can generalize it as :

```
Shape {
 appearance Appearance {
  material Material {
   emissiveColor x x x
   transparency x
  }
 }
 geometry IndexedLineSet {
  coord Coordinate {
   point [
```